

Homework 2

EE 999 — Introduction to Control Systems

Patrick Gordon • September 16, 2026

Due Wednesday, September 23, 2026

Problem 1.A.2.1 — What is a "free-body diagram"?

A free body diagram is a diagram that shows the forces acting on an object or set of objects.

Problem 1.A.2.2 — What are the two forms of Newton's law?

write the answer here

Problem 1.A.2.3 — For a structural process to be controlled, such as a robot arm, what is the meaning of "collocated control" and "Noncollocated control"?

write the answer here

Problem 1.A.2.4 — State Kirchhoff's current law.

Kirchhoff's current law states that current entering any given electrical node must equal current exiting a node.

Problem 1.A.2.5 — State Kirchhoff's voltage law.

Kirchhoff's voltage law states that the sum of all voltages around a closed circuit must equal zero.

Problem 1.B — Differential equation for mechanical system

The difference in velocities between m_1 and m_2 will decay because b_2 scales with velocity. Since there is no friction, the output $x_2(t)$ will not decay.